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Student Acceptance of AI-Driven Learning Analytics System

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Research Background

Artificial Intelligence (AI) is increasingly integrated into educational environments to support **personalized learning**, improve **efficiency**, and provide **data-driven insights**. However, the availability of AI technology does not automatically guarantee its adoption by students. Their willingness to use AI-driven systems may depend on whether the technology fits their needs, maintains their autonomy, is understandable, and can be used confidently. Therefore, this study examines the factors that influence students' acceptance and actual usage of **AI-Driven Learning Analytics Systems.**

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Research Problem

Although AI-driven learning analytics systems can support personalized and data-driven learning, their adoption among students is not automatic. Students may hesitate to use these systems when the technology does not fit their learning needs, limits their autonomy, or is difficult to understand. Therefore, this study examines the factors that influence students' intention to adopt and actually use AI-driven learning analytics systems.

System Compatibility

Does the AI-driven system fit students' learning needs, habits, and expectations?

User Autonomy

Can students maintain control and independence while using AI-based support?

Student Readiness

Do AI knowledge and technological confidence influence students' willingness to adopt the system?



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Research Objectives



This study aims to examine the factors that influence students' acceptance and actual usage of AI-Driven Learning Analytics Systems. The findings are expected to provide insights for developing more relevant and user-centered AI systems in educational environments.

Adoption Factors

Identify the factors that influence students' intention to adopt AI-driven learning analytics systems.

System Compatibility

Examine whether the system fits students' learning needs, habits, and expectations.

Actual Usage

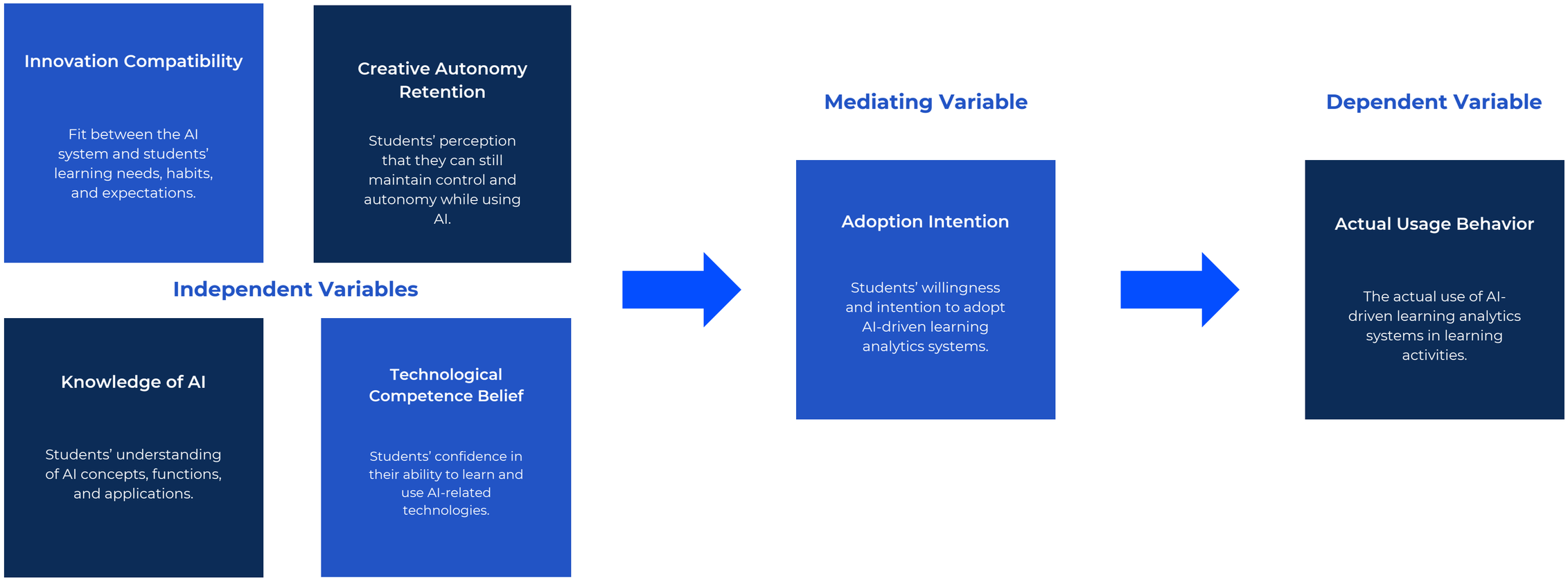
Analyze the effect of Adoption Intention on students' Actual Usage Behavior.

User-Centered Design

Provide insights for developing AI systems that are relevant, understandable, and supportive of student autonomy.

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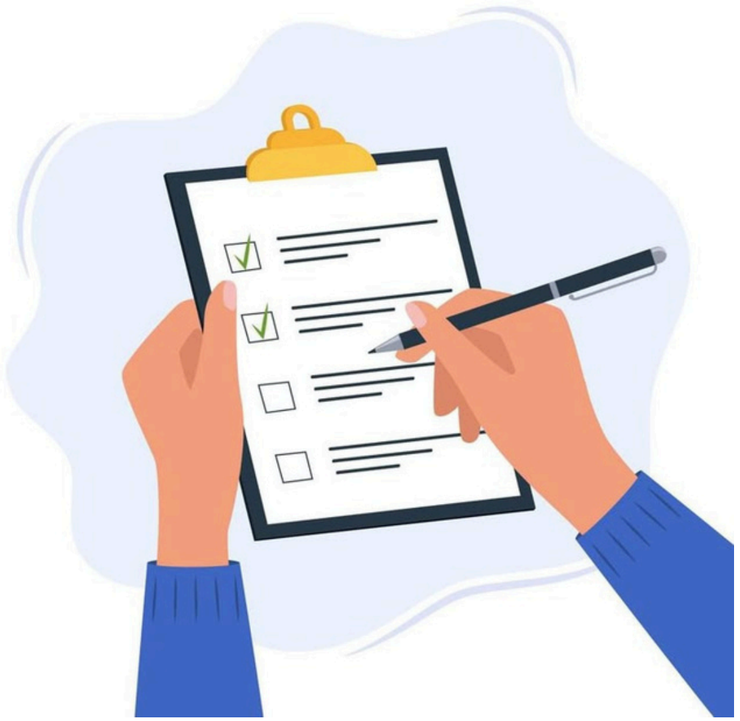
Research Model



This study examines the relationships between Innovation Compatibility, Creative Autonomy Retention, Knowledge of AI, and Technological Competence Belief toward Adoption Intention, as well as the effect of Adoption Intention on Actual Usage Behavior. In addition, Innovation Compatibility is also proposed to have a direct effect on Actual Usage Behavior.

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Research Methodology



This study applies a quantitative approach using a cross-sectional survey design. Data were collected through an online questionnaire and analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM) to examine the relationships between the proposed variables.

Quantitative Survey

6-Point Likert Scale

Data were collected through a self-administered online questionnaire using a six-point Likert scale from strongly disagree to strongly agree.

Respondents

244 Valid Responses

The study used purposive sampling. Respondents were university students familiar with AI-based tools in learning or creative activities.

Data Analysis

PLS-SEM

The data were analyzed using SmartPLS to evaluate the measurement model, structural model, and significance of the proposed relationships.

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Research Hypotheses

Innovation Compatibility positively influences **Adoption Intention.**	**Creative Autonomy Retention** positively influences **Adoption Intention.**	**Knowledge of AI** positively influences **Adoption Intention.**	**Technological Competence Belief** positively influences **Adoption Intention.**	**Adoption Intention** positively influences **Actual Usage Behavior.**
H1	H2	H3	H4	H5

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Respondent Profile

Data pre-processing was conducted before statistical analysis to improve data quality. From 251 initial responses, 7 responses were excluded due to duplicate entries and straight-lining patterns. Therefore, 244 valid responses were used for further analysis.

Initial Responses	Excluded Responses	Valid Responses
251	**7**	**244**
Total Responses	**Duplicate & Straight-Lining Responses**	**Responses Used for Analysis**



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Validity & Reliability Test

Outer Loadings

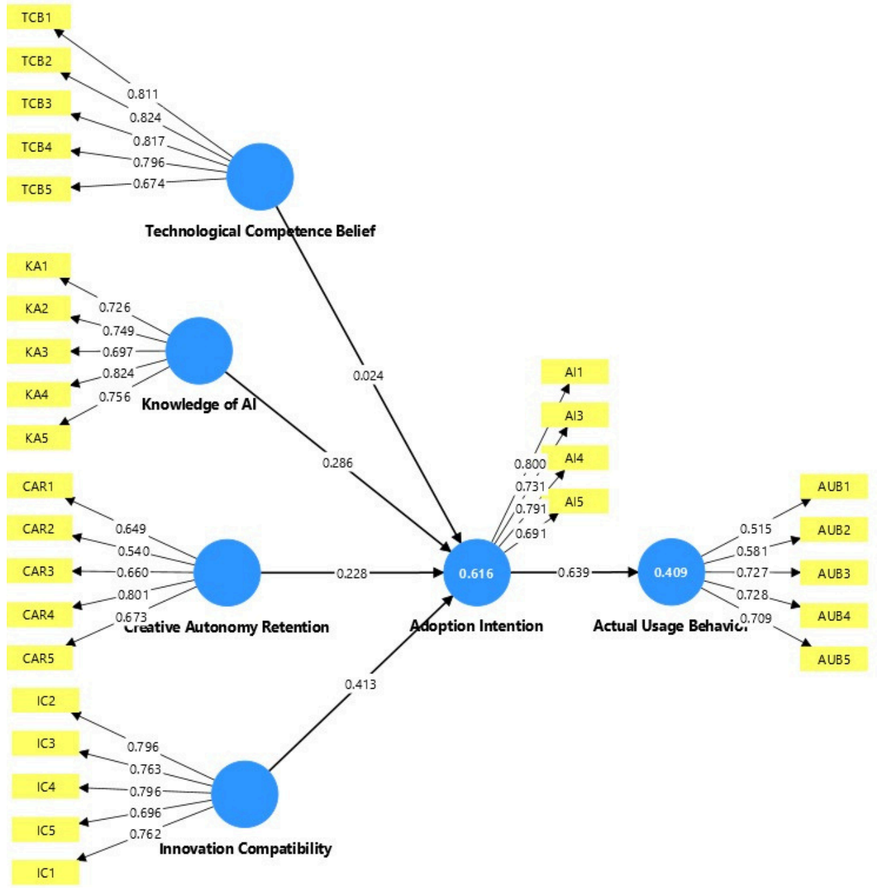
Most indicators were valid. Several indicators with values between 0.50 and 0.70 were retained after considering the overall quality of the measurement model.

Composite Reliability

All constructs achieved acceptable internal consistency, with Composite Reliability values ranging from 0.705 to 0.851.

AVE

Most constructs met the convergent validity criterion. However, Actual Usage Behavior and Creative Autonomy Retention still require further refinement.



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Hypothesis Testing Results

H1 — Supported

Innovation Compatibility → Adoption Intention

β = 0.413 | p-value = 0.000

H2 — Supported

Creative Autonomy Retention → Adoption Intention

β = 0.228 | p-value = 0.000

H3 — Supported

Knowledge of AI → Adoption Intention

β = 0.286 | p-value = 0.000

H4 — Supported

Technological Competence Belief → Adoption Intention

β = 0.024 | p-value = 0.024

H5 — Supported

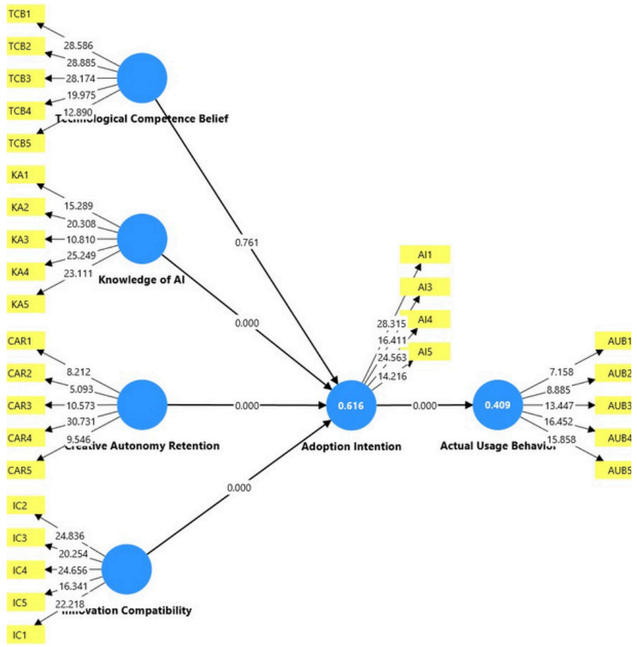
Adoption Intention → Actual Usage Behavior

β = 0.639 | p-value = 0.000

All five hypotheses were supported. Adoption Intention had the strongest effect on Actual Usage Behavior, while Innovation Compatibility was the strongest predictor of Adoption Intention.

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Key Findings



All five proposed hypotheses were supported. Adoption Intention had the strongest effect on Actual Usage Behavior. Among the predictors of Adoption Intention, Innovation Compatibility had the strongest influence, while Technological Competence Belief had the weakest but still significant effect.

Supported Hypotheses

5 / 5 Hypotheses Supported

All proposed relationships were statistically significant.

Path Coefficients (β)

Relationship	β
Innovation Compatibility → Adoption Intention	0.413
Creative Autonomy Retention → Adoption Intention	0.228
Knowledge of AI → Adoption Intention	0.286
Technological Competence Belief → Adoption Intention	0.024
Adoption Intention → Actual Usage Behavior	0.639

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Conclusion & Recommendations

This study concludes that students' acceptance of AI-Driven Learning Analytics Systems is influenced by Innovation Compatibility, Creative Autonomy Retention, Knowledge of AI, and Technological Competence Belief. Adoption Intention also has a significant effect on Actual Usage Behavior.

AI systems are more likely to be adopted when they are relevant, understandable, and user-centered.

System Fit

Design AI systems that match students' learning needs and habits.

Student Autonomy

Ensure students still feel in control when using AI-based support.

AI Literacy

Provide guidance or training to improve students' understanding of AI.

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We are ready to assist you